
EECS 16A Touchscreen 3B

****Insert Lab Staff Names****

Semester Outline



Imaging
Module



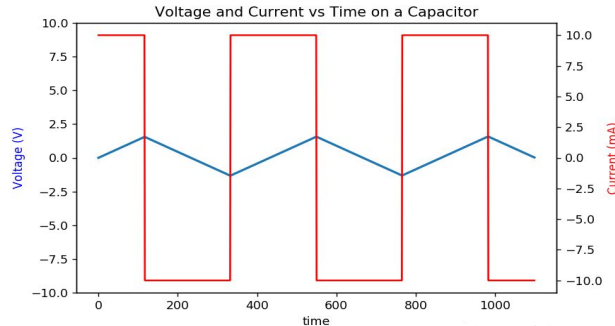
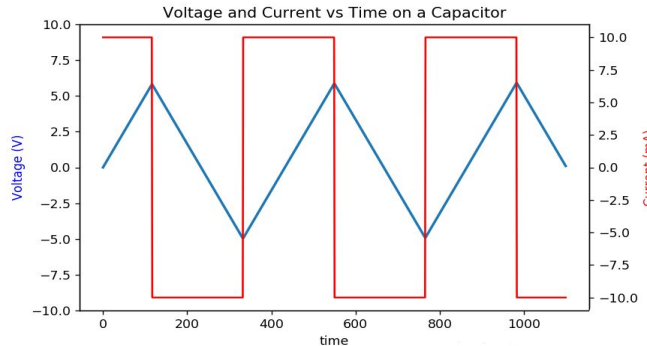
Touchscreen
Module



Acoustic
Positioning
Module

Last time: Touch 3A

- Simulated a touch-sensing circuit
 - Current source onto capacitor gave: $V(t) = \frac{I}{C}t + V_0$
 - Periodically charging and discharging (square wave current) gave a triangular shaped waveform
- **What changed between touch and no touch?**
 - Touch = Add capacitors (increase capacitance)
 - Increase in capacitance => Decrease in voltage
 - Can see this change with a comparator and an LED!



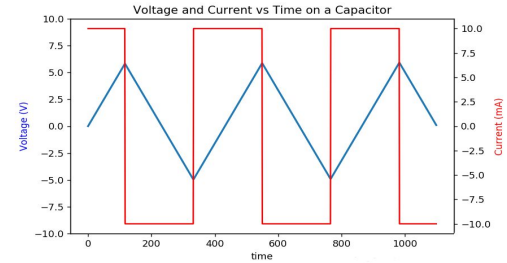
Last time: Touch 3A

- Problem: **we don't have ideal square current sources**
 - We need another way to implement last lab's voltage waveforms (the triangle wave output)

ELECTRICAL SYSTEM
WITH
VOLTAGE/CURRENT
OUTPUT



TOUCH 3A
CIRCUIT



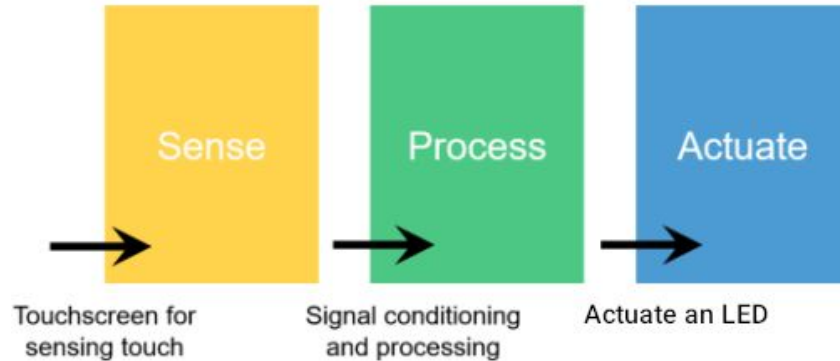
- How do we go about creating a similar system that still fits our model?

This week: Touch 3B

- Explore an alternative to ideal current sources
 - Use our new (and proven) op amp skills to build an equivalent circuit
- Build a complete system that will detect touch and have an actuation component (LED)

Electronic Systems: A review

- Sensing is only a part of a complete system. Most systems perform 3 tasks:
 - Sense (Physical to Electrical)
 - Process (Signal Conditioning)
 - Actuate (Electrical to Physical)



Building a Current Source (Note 20)

- Need a circuit that **outputs a constant current regardless of voltage across**
- What we have:
 - Voltage sources
 - $V = IR$ relationship for resistors
 - Note 20's guidance

20.3 Design Example: “Almost” current source

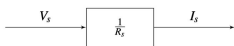
In this section, we will use resistors, voltage sources, and op amps to build a current source. We want a source that can be used as the current source in the countdown timer above. Let's follow our design procedure.

Step 1 (Specification): First we restate the problem clearly: Build a current source that outputs constant current, I_s , regardless of the voltage across it.

Step 2 (Strategy): We can use a voltage source, but we need a way to transform the voltage into a constant current. Ohm's law tell us

$$I = \frac{V}{R},$$

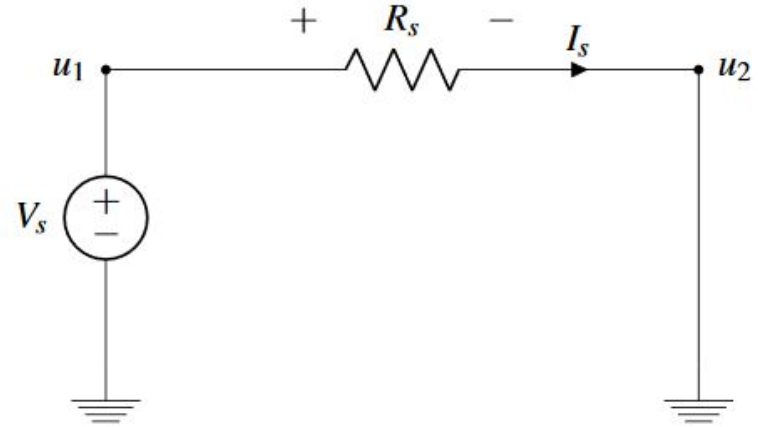
so maybe we can use a resistor to convert the voltage source output, V_s , into a constant current, I_s . This is summarized in the following block diagram:



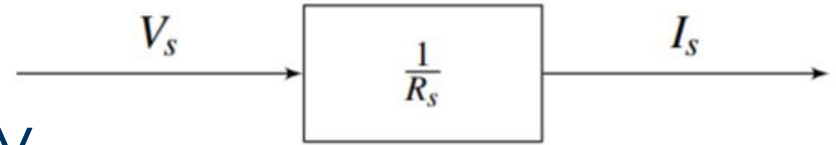
Step 3 (Implementation): Based on our strategy from Step 2, we make our first attempt to build the current source. Let's take a voltage source and connect it to a resistor:

First Attempt at a Current Source

- If we have a voltage source and a resistor then we can create a “current source”

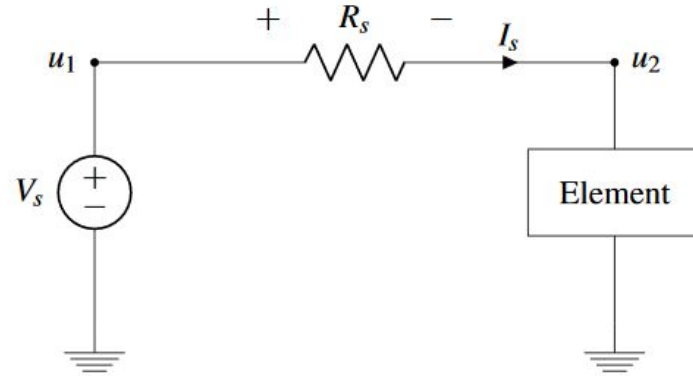


- The current, $I_s = \frac{V_s}{R_s}$
since the other side (u_2) is 0V



First Attempt Evaluation

- What happens when we attach a load?
- Assume that the element is a resistor of value R_L
- **Does this work?**

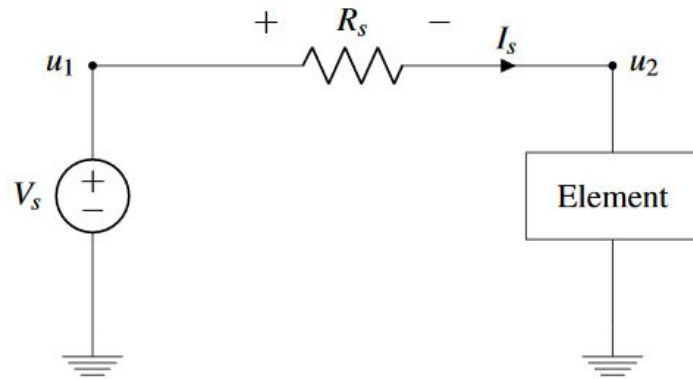


NOPE, it changes the current

$$I = \frac{V_S}{R_S + R_L} \neq \frac{V_S}{R_S}$$

Try Again

- The issue here is that we had $I_s = \frac{V_s - 0}{R_s}$
- But a load made it so R_s isn't connected to 0 on the other side
- We need to set the u_2 node voltage to 0 for this to work
- **Do you know anything that can force nodes to 0V?**

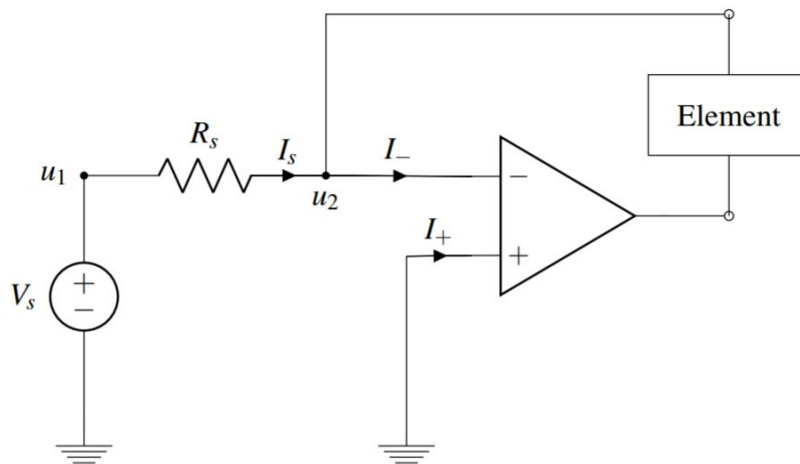


Note 20: An “almost” Current Source

- We can use an **op amp**!

Recall, the **GOLDEN RULES**:

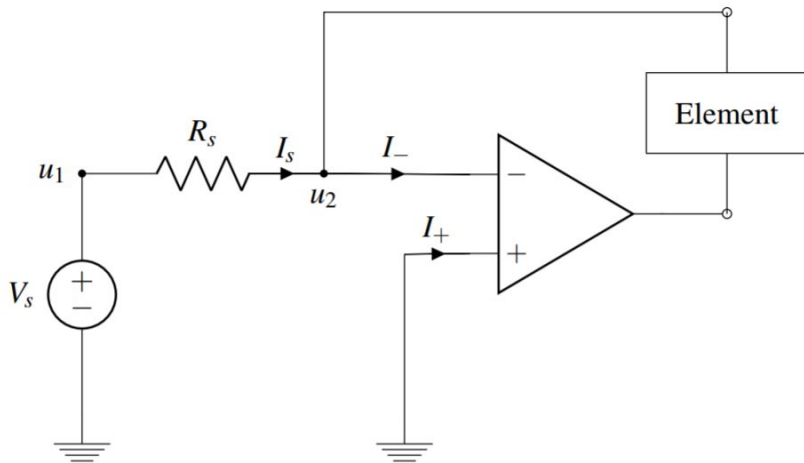
- GR #1: No current going into the op amp
- GR #2: $u_+ = u_-$, so let's make one of them 0V
 - **What must be true for this to hold?**



Note 20: An “almost” Current Source

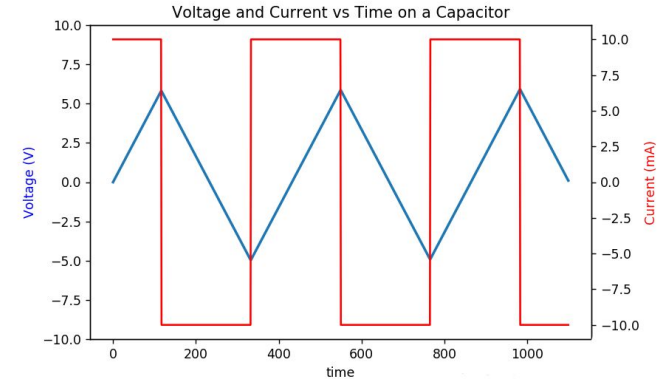
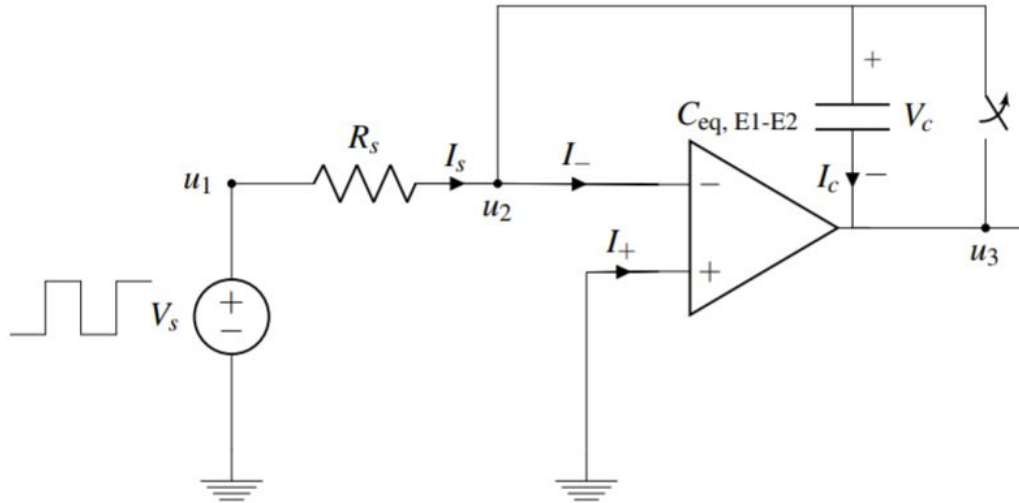
- Since we are in **negative feedback**, $u_2 = 0V$
 - $I_S = \frac{V_S - 0}{R_S}$
- All current will go to the element, since $I_- = 0$

By KCL, $I_S = I_{element} + I_- = I_{element}$



Sensing a Completion

- Hook up our capacitive touchscreen from Touch 3A
- We get a constant current through the capacitor
- **What's the output of this circuit?**



Note 20: An “almost” Current Source

- Constant current is cool, but we want periodic current to discharge the cap.
- What if we periodically switch voltage?

$$I_S = \frac{V_s}{R} \longrightarrow I_S = \frac{-V_s}{R}$$

An Alternate Viewpoint

- Note that the output of this circuit is
- It's also an integral, just like last time!
$$V_{out} = -\frac{1}{R_s C} \int_0^t V_s dt$$
 - New circuit is an “almost current source” or just trading current for voltage.
- **We're now integrating a constant voltage instead of a current**, but the net result is the same as last time
- We traded one type of input for another!
- Variable voltage sources do exist, so this is good.
What are they like though?

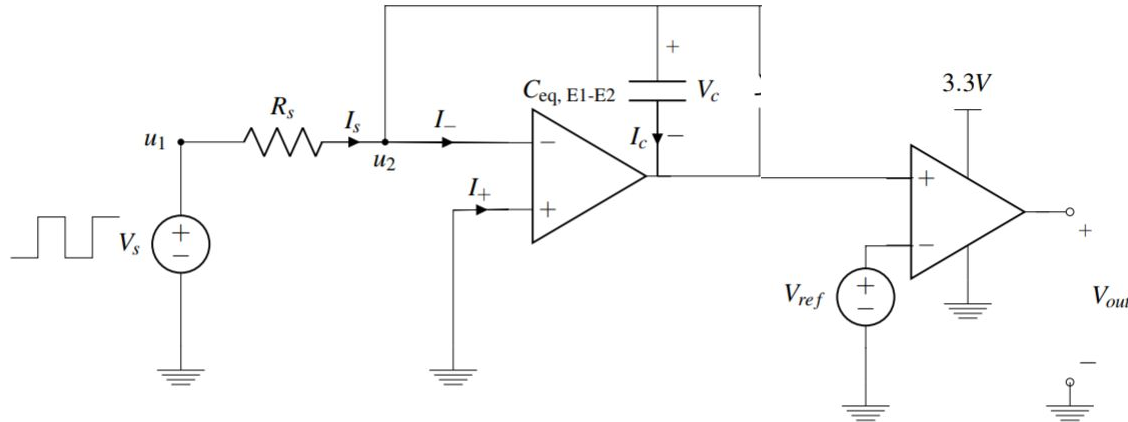
What's our (new) input?

- Function generator
- Can create different waves
- Treat it as a non-constant voltage source
- Now we can make our “almost current source”!

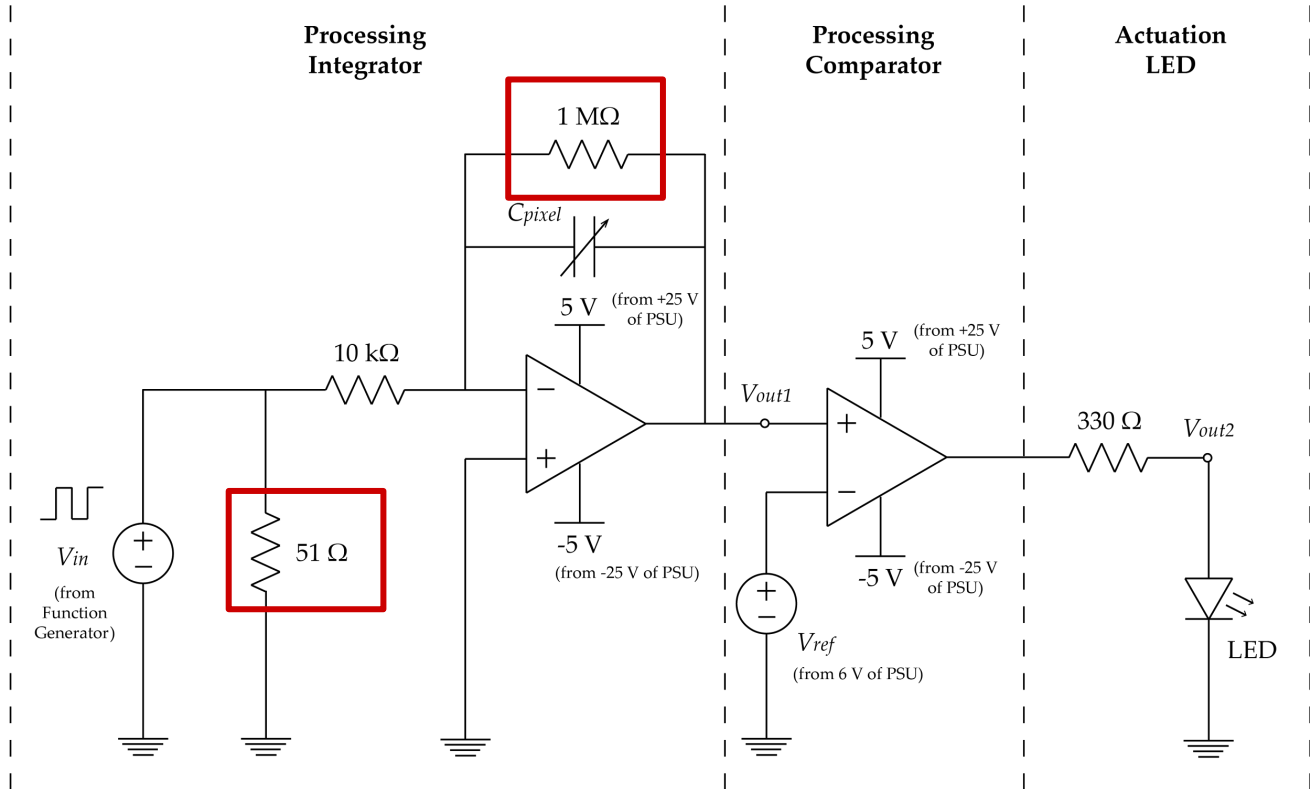


Processing the rest of our system

- Our circuit behaves as intended
- We can feed the new signal into our comparator circuit from last time



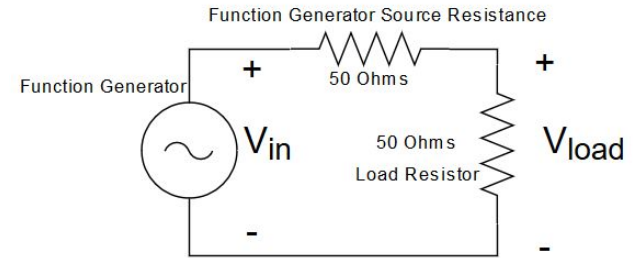
Our real-world circuit



51 ohm? Voltage Dividers!

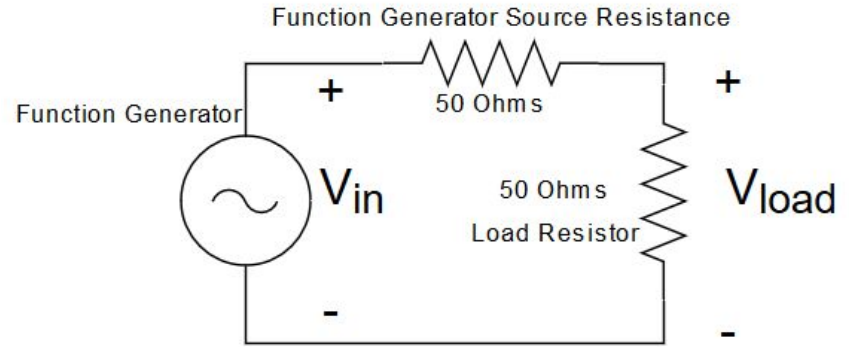
- The function generator has a 50 Ohm source resistance
- Our function generator also assumes a 50 Ohm load is attached (just because).
 - **What's the voltage you get across this load?**

If you attach a 50 Ohm load, then the load only gets $\frac{1}{2}$ of V_{in} applied



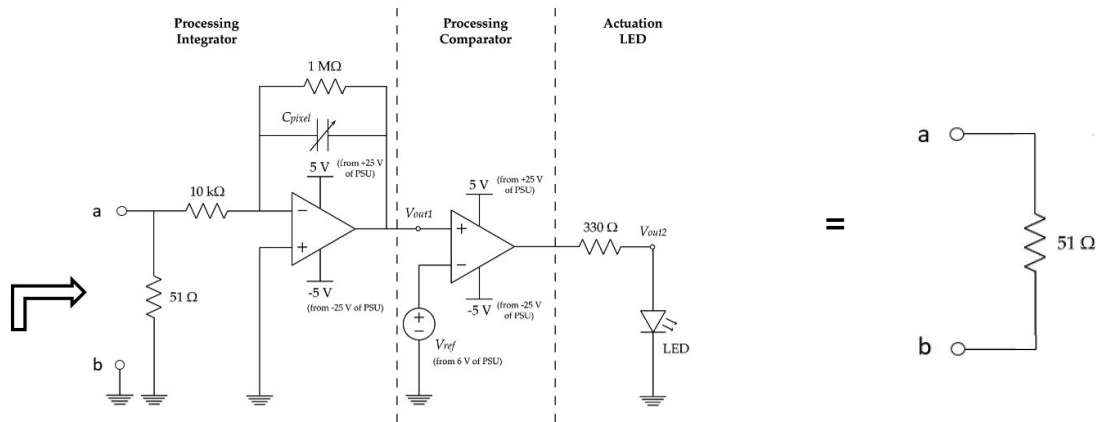
51 ohm? Voltage Dividers!

- The function generator will automatically double its output voltage (V_{in}) so that the voltage across the load (V_{load}) is what you would expect after it is halved



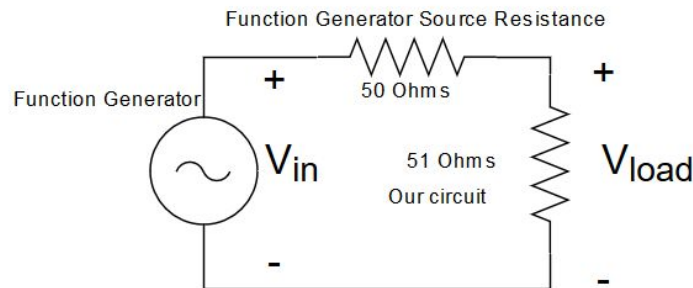
What does the 51 ohm do?

- Compute the thevenin resistance of our circuit from the input port
 - It's about 51 Ohms
- Our circuit (from the input) looks like a 51 Ohm resistor

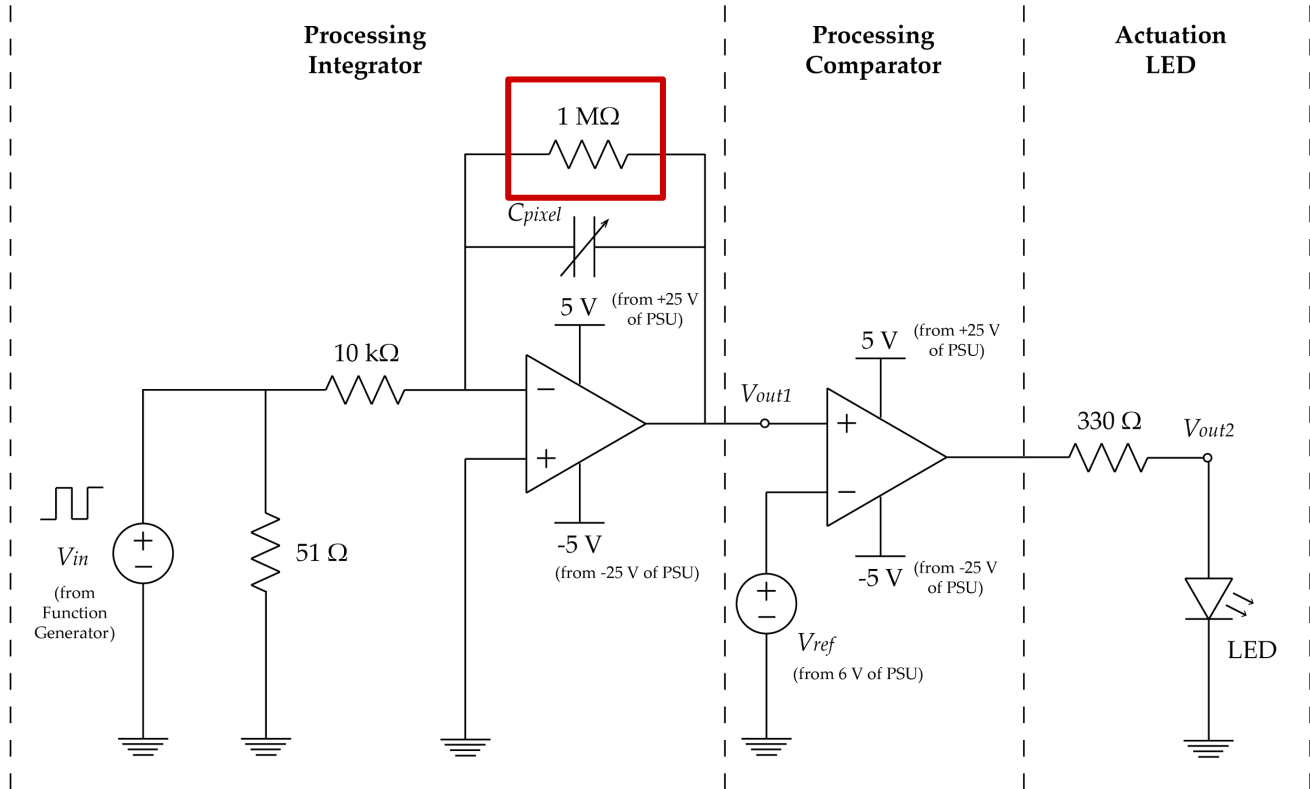


What does the 51 ohm do?

- Our circuit looks like a 51 ohm load with respect to the input, so the function generator is happy!
- (Note: 50 Ohm resistors basically don't exist so we use 51 because it's the next closest value)



Our real-world circuit



What does the 1 megaohm do?

- It's a little out of scope
- It ensures that the circuit is always in negative feedback
 - Since it's 1 million Ohms it draws almost 0 current, and thus doesn't really affect our analysis
- If it was not there, the Capacitor acts as an open during constant voltage, so there is no feedback

Taking the limit

- Didn't you say the capacitive touchscreen is way better than resistive? Why do we only have one touch point instead of nine?

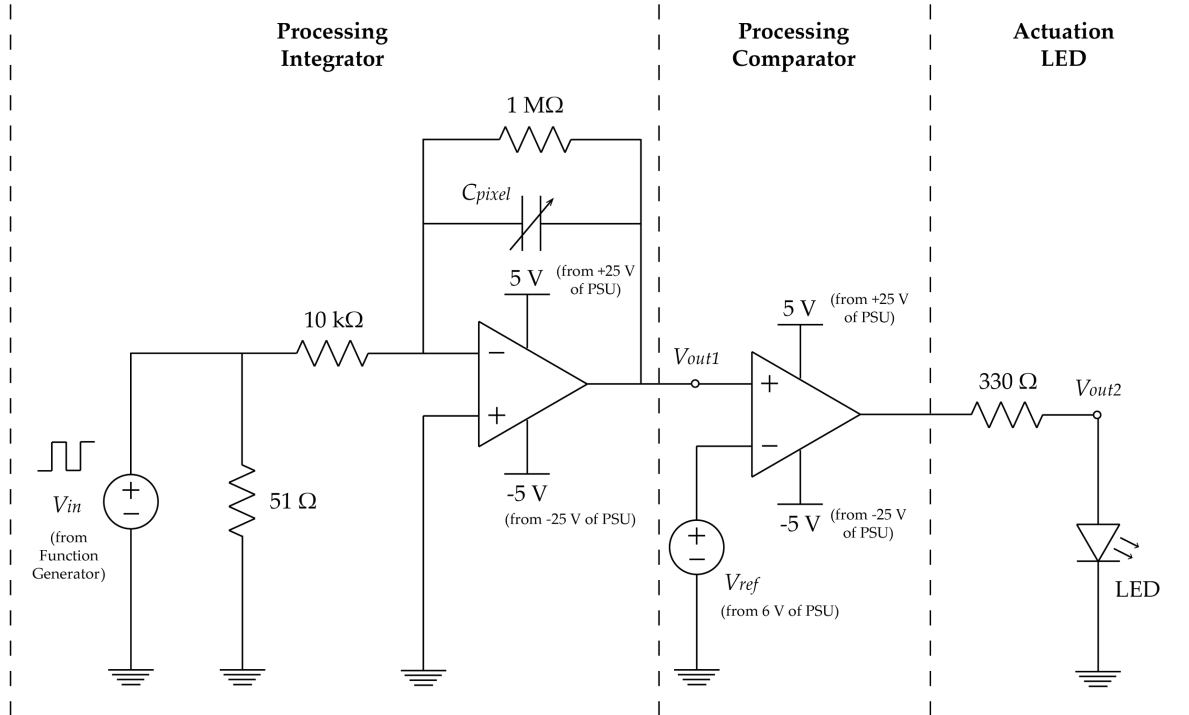
Taking the limit

- Note that this isn't dependent on voltage dividers at all, only on if you are locally touching the capacitor
- **How to add more touch points?**
 - Duplicate the entire circuit and put them next to each other. Each “touchscreen” we have built is actually a pixel!
- They're independent, so the more you add the more points you can sense

Taking the limit

- Make the capacitors really small, put them in the size of a screen
- Thousands of these sensing circuits can be made incredibly small
 - (less than 4mm x 4mm)
- Put a thousand of these and you can recognize 1000 different touch points
- No moving parts, much better (and more accurate) than the resistive touchscreen

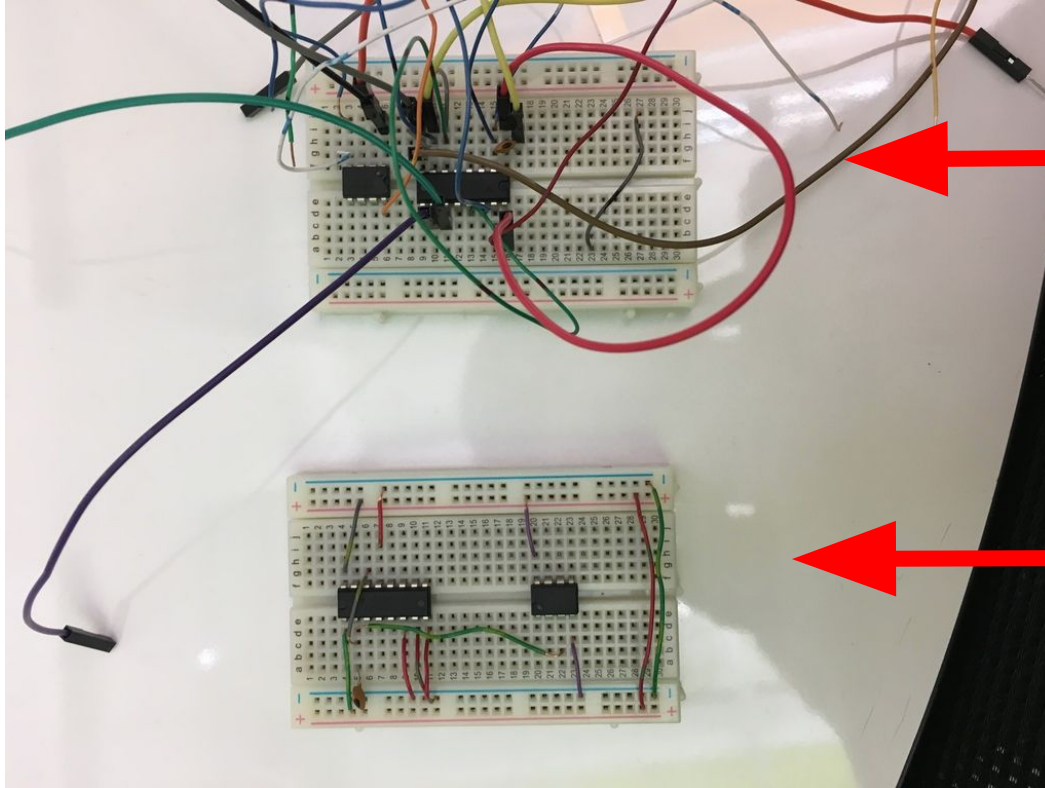
That's it!



Quick note

- Planar wiring required
- We can and will refuse to help you fix your circuit if it's too messy
 - Use the breadboarding wires at the TA desk and the wire strippers at your stations
 - Cut wires and resistors to be as short as you can and have them still work.

Why is planar wiring required?

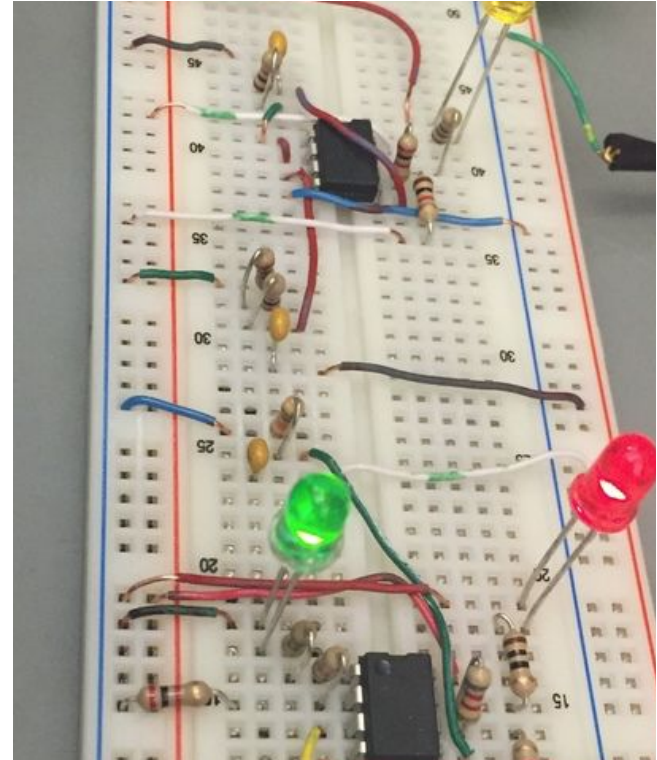


1.5 Hour to debug;
Falls apart easily

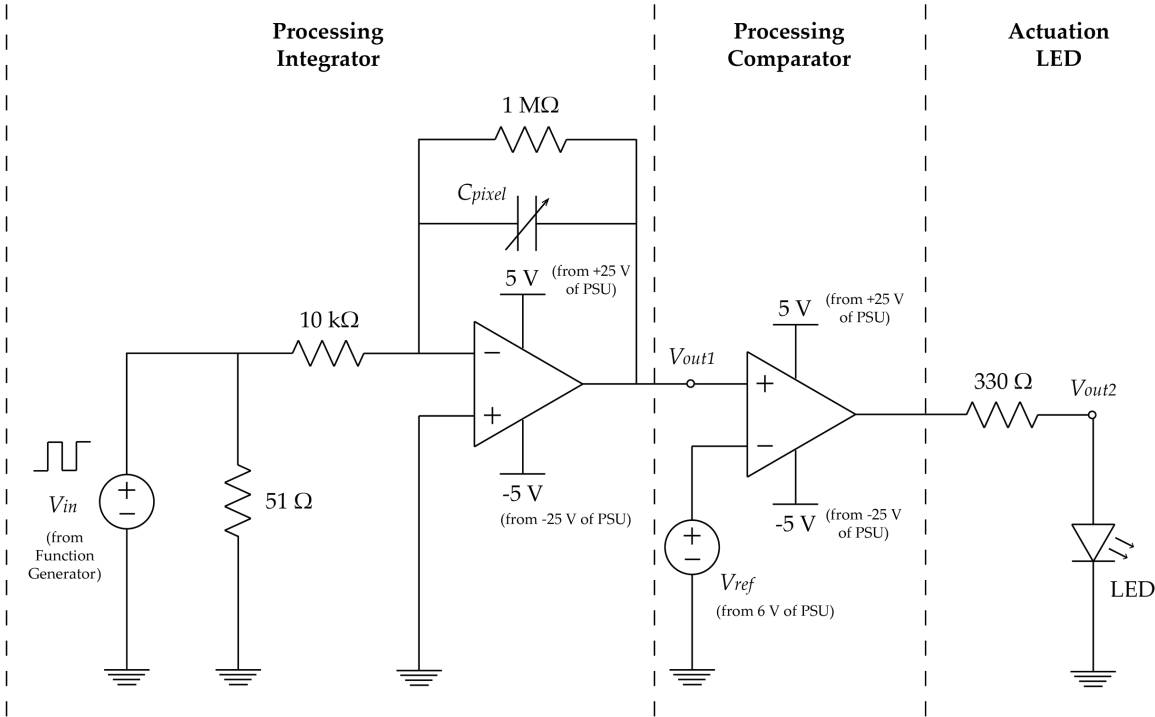
5 seconds to debug;
Practically 2D;
Lasts a lifetime

Keep your circuits neat!

- **Cut wires to correct lengths.**
- Place op amp across the middle of your breadboard (should already be there).
- If circuit is not neat, will not debug until it is.
- Get Started!



And that's it!



R	Band Colors
51 Ω	Gr Br Bl G
10 k Ω	Br Bl O G
1 M Ω	Br Bl Gr G
330 Ω	O O Br G